



SIM32ELA

Hardware Design

GNSS Module

SIMCom Wireless Solutions Limited

SIMCom Headquarters Building, Building 3, No. 289 Linhong
Road, Changning District, Shanghai P.R. China

Tel: 86-21-31575100

support@simcom.com

www.simcom.com

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SIMCom Wireless Solutions

SIMCom Headquarters Building, Building 3, No. 289 Linhong Road, Changning District, Shanghai P.R.China

Tel: +86 21 31575100

Email: simcom@simcom.com

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2023-02-13	V1.01	1. Update Table 1 A-GPS TTFF(EPO in flash mode) data 2. Modify the Table 6 RTC mode power supply state 3. Refine the description of the 3.3 EINT_IN signal 4. Update Figure 18 reflow furnace temperature curve	QIU LEI

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1 Introduction

This document describes the hardware interface of the SIMCom module SIM32ELA, which is a high performance and reliable GNSS module. It is a GNSS module integrated with GPS & GLONASS & BEIDOU & Galileo & QZSS system in a LCC type with AIROHA's high sensitivity navigation engine, which allows customer to achieve industrial high level sensitivity, accuracy, and Time-to-First-Fix (TTFF) with lower power consumption..

SIM32ELA provides simultaneous GPS, BEIDOU, GLONASS, Galileo and QZSS open service L1 reception capability. With 33 tracking channels and 99 acquisition channels, SIM32ELA can acquire and track any mix of multiple satellite signals. Combining advanced AGPS called EASY™ (Embedded Assist System) with proven AlwaysLocate™ technology, SIM32ELA achieves the highest performance and fully meets the industrial standard..

Key Features

The module provides complete signal processing from antenna input to host port in NMEA messages. SIM32ELA contains a built-in antenna, so you can choose to connect an external antenna or use this built-in antenna. The module requires 2.8V to 4.3V power supply, which gives customers plenty of choices for the application circuit. The host port is configurable to UART. Host data and I/O signal levels are 2.8V CMOS compatible.

- GPS, BEIDOU, GLONASS, Galileo and QZSS open service L1 reception capability
- 33tracking/99 acquisition-channel GNSS receiver
- Small footprint: 9.6x 14 x 2.15mm, 31-pin LCC package
- Support Jamming Removing
- Max fixed update rate up to 10 HZ¹
- Advanced software features
 1. Always locate advanced location awareness technology
 2. EPO orbit prediction
 3. EASY self-generated orbit prediction
 4. SBAS ranging (WAAS, EGNOS, GAGAN, MSAS)
- Pulse-per-second (PPS) GPS time reference
 1. Adjustable duty cycle
 2. Typical accuracy: +/- 10ns
- Interface
 1. UART²
- Operating temperature: -40 ~ +85°C
- RoHS compliant

NOTE

1. Default is 1 HZ;
2. UART0 for output NMEA and download, UART1 for RTCM function.

1.1 SIM32ELA Functional Diagram

The following figure shows a functional diagram of the SIM32ELA and illustrates the mainly functional parts:

- The main chip
- SAW filter
- The antenna interface
- The communication interface
- The control signals
- Internal Antenna(Internal LNA)

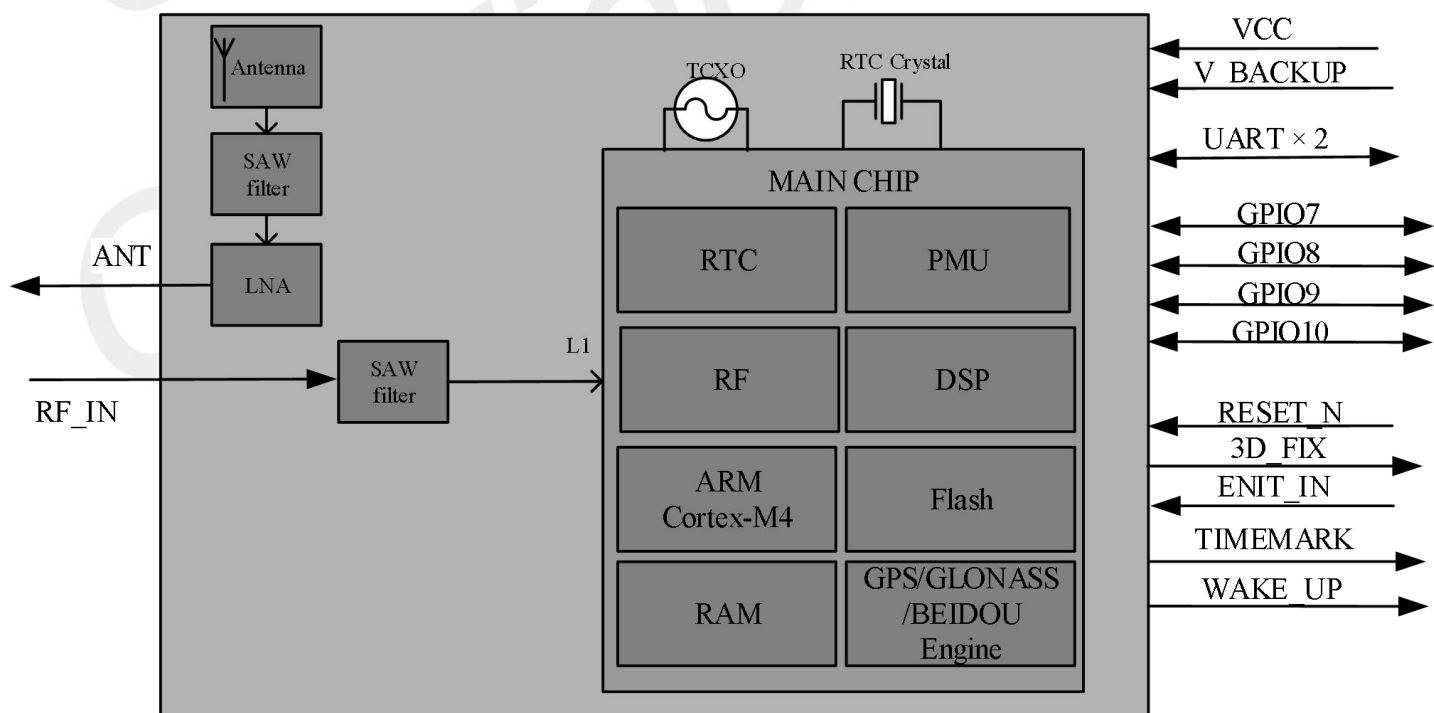


Figure 1: SIM32ELA functional diagram

1.1.1 GNSS Performance

Table 1: GNSS Performance

Parameter	Description	Performance			
		Min	Type	Max	Unit
Horizontal Position Accuracy ¹	Automatic position		1		m
Velocity Accuracy ²	Without Aid		0.1		m/s
	DGPS		0.05		m/s
Timing Accuracy			10		nS
Dynamic Performance	Maximum Altitude			18000	m
	Maximum Velocity			500	m/s
	Maximum Acceleration			4	G
TTFF with GPS (L1) and GLONASS and BEIDOU mode	Hot start		0.46		S
	Warm start		15.4		S
	Cold start		15.46		S
A-GPS TTFF(EPO in flash mode)	Hot start		0.61		S
	Warm start		20.45		S
	Cold start		20.77		S
Sensitivity with GPS (L1) and GLONASS and BEIDOU mode	Autonomous acquisition(cold start)		-148.5		dBm
	Re-acquisition		-158.5		dBm
	Tracking		-167		dBm
Receiver	Channels		L1:75		
	Update rate		1	10	Hz
	Tracking L1, CA Code				
	Protocol support NMEA,PAIR				
Power consumption With GPS (L1) and GLONASS and BEIDOU mode ⁴	Acquisition		13		mA
	Continuous tracking		13		mA
	Sleep current(with uart sleep)		200		uA
	RTC current		15		uA

NOTE

1. 50% 24hr static, -130dBm;
2. 50% at 30m/s;
3. GPS signal level: -130dBm; GLONASS signal level: -130dBm; GALILEO signal level: -130dBm; BEIDOU signal level: -130dBm;
4. Power supply 3.3V;

1.1.2 General features

Table 2: General features

Parameters		Value
Supply voltage VCC		+2.8V~4.3V typical:3.3V
Supply voltage ripple VCC		54 mV(RMS) max @ f = 0~3MHz 15 mV(RMS) max @ f > 3 MHz
Power consumption(acquisition)		18mA.@ VCC=3.3 V
Power consumption(sleep)		200uA.@ VCC=3.3 V
Power consumption(RTC)		15uA.@ V_BACKUP=2.8V
Storage temperature		-45°C~+95°C
Operating temperature		-40°C~+85°C ¹
DVDDIO=2.8V I/O signal levels(V)	V _{IL}	-0.3V~0.7V
	V _{IH}	1.75V~3.1V
	V _{OL}	-0.3V~0.35V
	V _{OH}	2.1V~3.1V
DVDDIO=1.8V I/O signal levels(V)	V _{IL}	-0.3V~0.45V
	V _{IH}	1.35V~2.1V
	V _{OL}	-0.3V~0.27V
	V _{OH}	1.53V~1.98V
I/O output sink/source capability		+/- 8mA max
I/O input leakage		+/- 5 uA max
Host port		UART0
Other port		UART1
Serial port protocol (UART)		NMEA; 8 bits, no parity, 1 stop bit; 115200 baudrate (configurable)
TIMEMARK output (1PPS)		PPS 1 pulse per second, synchronized at rising edge, The rising edge of the pulse is aligned with UTC seconds, with an accuracy of about 10nS pulse length 100ms

2 Package Information

2.1 Pin out Diagram

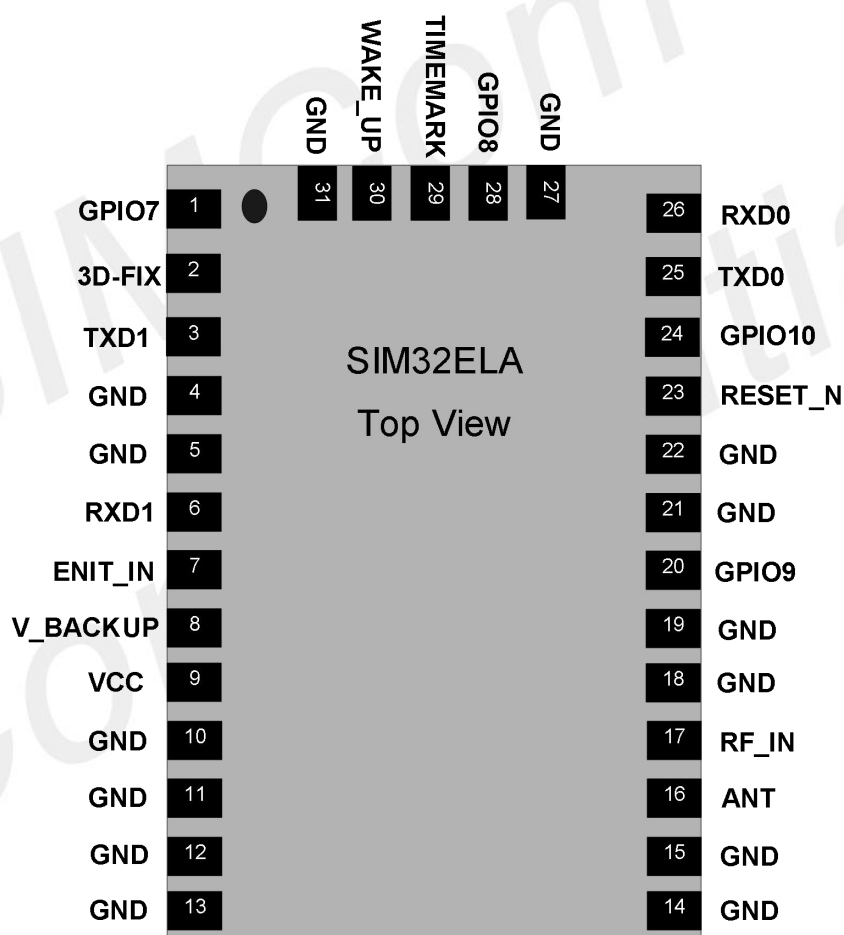


Figure 2: SIM32ELA pin diagram (Top view)

2.1.1 Pin Description

Table 3: I/O parameter definition

Type	Description
AI	Analog input
AO	Analog output
DI	Digital input
DO	Digital output
PI	Power input
PO	Power output

Table 4: Pin description

Pin name	Pin number	I/O	Description	Comment
Power supply				
VCC	9	PI	Main power input, which will be used to power the baseband and RF section internally.	Provide clean and stable power source to this pin. Add a 4.7uF capacitor to this pin for decoupling.
WAKE_UP	30	PO	1.8V power output. High level indicating to host that the module is active.	If unused, keep open.
V_BACKUP	8	PI	The backup battery input power supply for RTC	It must be connected
GND	4,5,10,11,12,13,14,15,18,19,21,22,27,31		Ground	GND
Host port interface				
TXD0	25	DO	NMEA serial output/ input	2.8V power domain, If unused, keep open.
RXD0	26	DI		
TXD1	3	DO	Default RTCM function	
RXD1	6	DI		
GPIOs				
3D-FIX	2	DO	Location flag. After the location is successful, the output is high	2.8V power domain, If unused, keep open.
TIMEMARK	29	DO	TIMEMARK outputs timing pulse related to receiver time	2.8V power domain, If unused, keep open.
RESET_N	23	DI	Reset the module, active low	
ENIT_IN	7	DI	Wake up SIM32ELA, low	2.8V power domain,

			active , default pull high	If unused, keep open.
GPIO7	1	DI/DO	General purpose input/output	2.8V power domain, If unused, keep open..
GPIO9	20	DI/DO	General purpose input/output	2.8V power domain, If unused, keep open..
GPIO10	24	DI/DO	General purpose input/output	2.8V power domain, If unused, keep open..
GPIO8	28	DI/DO	General purpose input/output	2.8V power domain, If unused, keep open..
RF interface				
RF_IN	17	AI	Radio antenna connection	Impedence must be controlled to 50Ω.
ANT	16	AO	Built-in antenna output	Connect to RF_IN, If you do not use the built-in antenna, keep open

2.1.2 Package Dimensions

Following figure shows the Mechanical dimensions of SIM32ELA (top view, side view and bottom view).

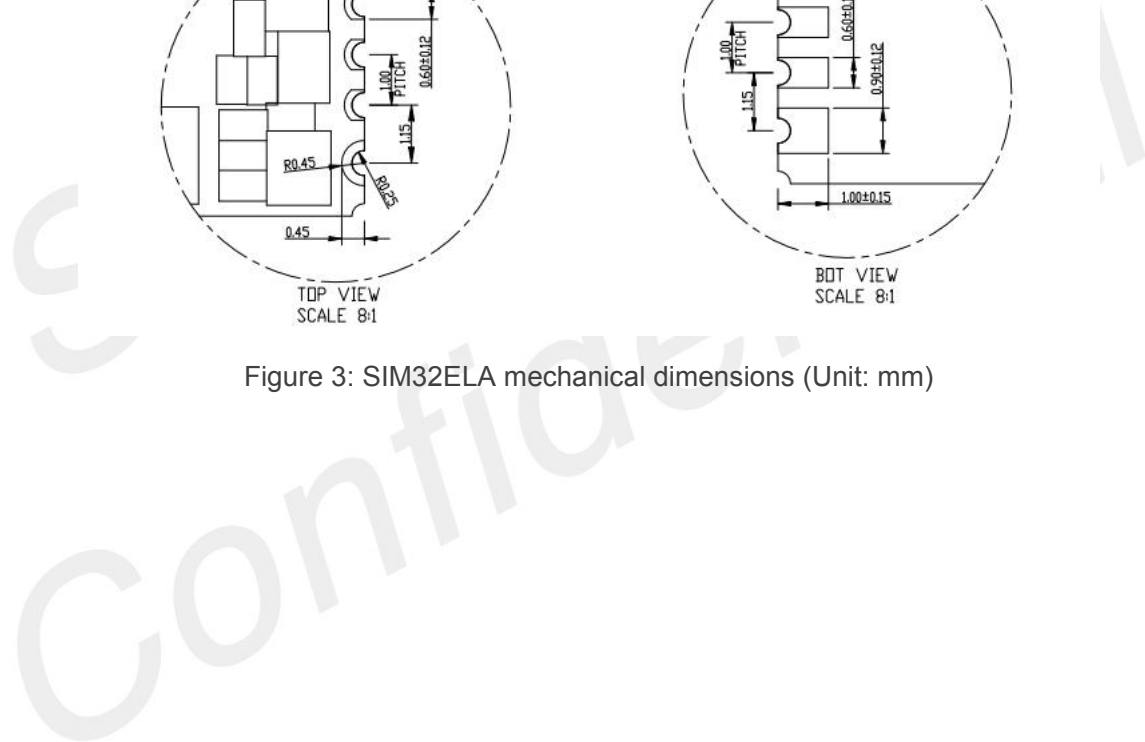


Figure 3: SIM32ELA mechanical dimensions (Unit: mm)

3 Application Interface

3.1 Power Management

3.1.1 Power Input

VCC supply requirements

The power supply range of SIM32ELA is from 2.8V to 4.3V, typical 3.3V. The power supply should be able to provide sufficient current up to 100mA. SIM32ELA positioning modules require a stable power supply, consider the following points:

- The power cable must be wide enough.
- VCC supply needs to add a 4.7uF and 100nF multi-layer ceramic chip (MLCC) capacitors with low ESR in high frequency band, which can be used for EMC performance .
- The ripple of the VCC supply cannot be higher than 15mV.
- VCC supply needs a ESD and surge protection.

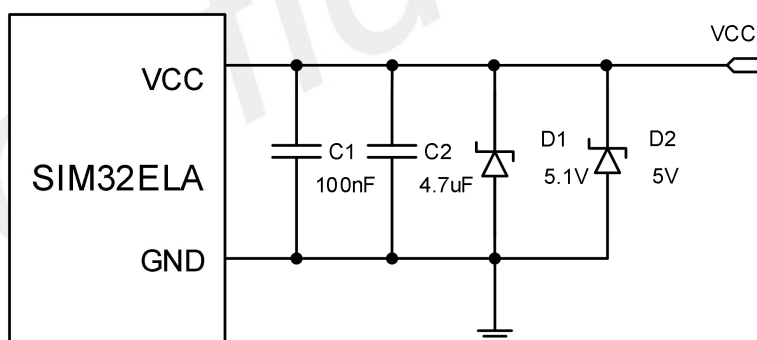


Figure 4: Power supply reference circuit

NOTE

1. C1 and C2 are multi-layer ceramic chip (MLCC) capacitors with low ESR in high frequency band, which can be used for EMC performance.
2. D2 is used for ESD protection and D1 is used for surge protection.

Table 5: Recommended D1 and D2 list

No.	Manufacturer	Part number	VRWM	Package	Ref. Designator
1	JCET	ESDBW5V0A1	5V	DFN1006-2L	D2
2	WAYON	WS05DPF-B	5V	DFN1006-2L	
3	LRC	LEDZ5.1BT1G	5.1V	SOD-523	D1
4	Prisemi	PZ5D4V2H	5.1V	SOD-523	

Backup supply requirements

Using valid time and SIM32ELA orbit data at startup will improve GPS/GNSS performance. Enable hot start. In order to take advantage of these functions, connect a power source (such as a battery, etc.) to V_BACKUP to continue supplying power to the backup domain in the event of a power failure in the VCC after sending a command to enter the RTC mode.

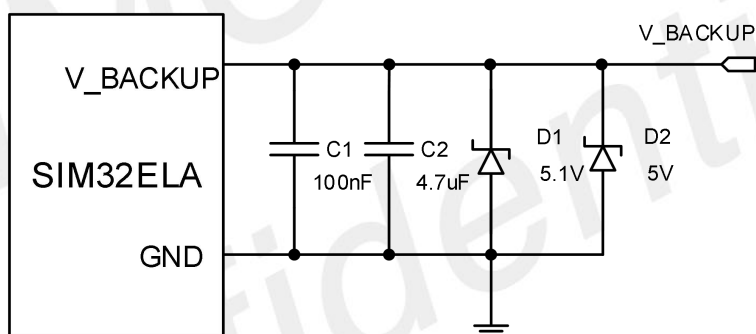


Figure 5: V_BACKUP reference circuit

3.1.2 Starting SIM32ELA

When power is first applied, SIM32ELA goes into operation mode.

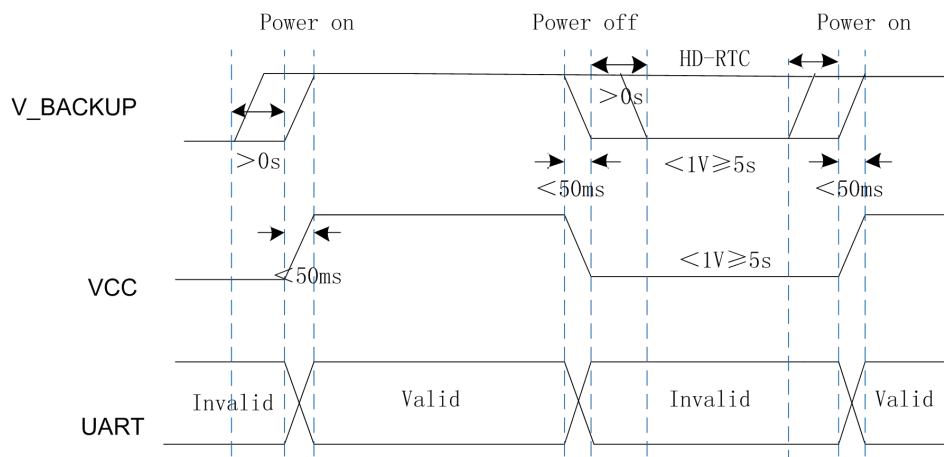


Figure 6: Timing sequence of module power-on and power-off

3.1.3 Verification of SIM32ELA Start

System activity indication depends upon the chosen serial interface:

When it is activated, SIM32ELA will output messages at the selected UART speed, and message types. The default baud rate is 115200bps.

3.1.4 Power Saving Modes

SIM32ELA supports power saving modes for reducing average power consumption like sleep mode, RTC mode.

- Sleep mode: In this mode the receiver stays at full on power state. This mode can be woken up by the host by sending external interrupt.
- RTC mode: Software on host side to send the command through the communication interface to into the soft RTC mode, wake up by RESET_N.

NOTE

The modes mentioned above are operated by PAIR commands, users can refer to “SIM32ELA Series_NMEA Message_User Guide” for more information.

3.1.5 Operating Mode

Table 6: Power supply and clock state according to operation mode

Mode	VCC	V_BACKUP	Internal LDO	Main clock	RTC clock
Full on	on	on	on	on	on
Sleep mode	on	on	on	off	on
RTC mode	off	on	off	off	on

Full on Mode

The module will enter full on mode after first power up with factory configuration settings. Power consumption will vary depending on the amount of satellite acquisitions and number of satellites in track. This mode is also referenced as full on, full power or navigation mode.

First positioning (different CN value, different positioning time), can download complete ephemeris data 15 minutes after positioning.

Navigation is available and any configuration settings are valid as long as the VCC power supply is active. When the power supply is off, settings are reset to factory configuration and receiver performs a cold start on next power up.

Sleep Mode

Sleep mode means a low quiescent power state(2mA type), non-volatile RTC, and backup RAM block is powered on. Other internal blocks like digital baseband and RF are internally sleeping. The PMU is changed to low power mode, The power supply input VCC shall be kept active all the time, even during sleep mode. If lower power consumption is required, you can send \$PAIR382,0*2F command to turn on the UART sleep function(200uA type).

UART sleep :

- Enter sleep mode:

Enter sleep mode is controlled by UART interface, send \$PAIR003*39\r\n, SIM32ELA will return \$PAIR001,003,1*39\r\n,\$PAIR001,003,0*38\r\n, then enter sleep mode.

- Exit sleep mode:

Pull down “ENIT_IN” pin 10ms, in the process of pulling down ENIT_IN 10ms, SIM32ELA will return \$PAIR012*39, and send \$PAIR002*38\r\n to exit sleep mode within 100ms after returning \$PAIR012*39. The module will return \$PAIR001,002,1*38\r\n, \$PAIR001,002,0*39\r\n, then exit sleep mode.

UART does not sleep:

- Enter sleep mode:

Enter sleep mode is controlled by UART interface, send \$PAIR003*39\r\n, SIM32ELA will return \$PAIR001,003,1*39\r\n,\$PAIR001,003,0*38\r\n, then enter sleep mode.

- Exit sleep mode:

Send \$PAIR002*38\r\n to exit sleep mode, the module will return \$PAIR001,002,1*38\r\n, \$PAIR001,002,0*39\r\n, then exit sleep mode.

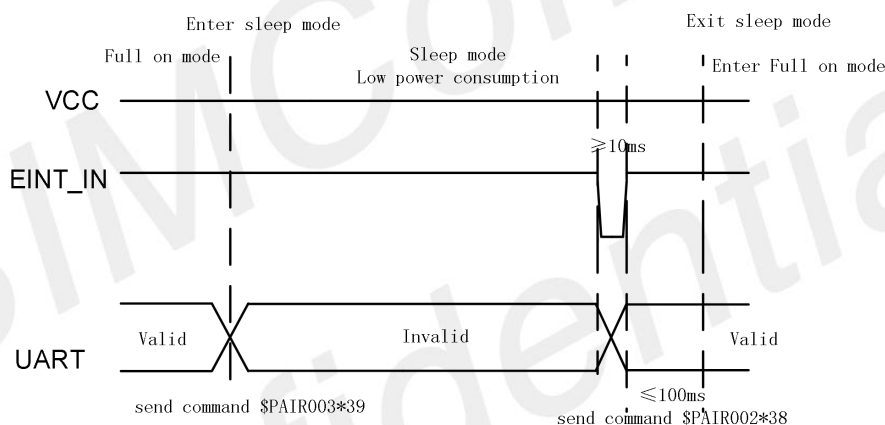


Figure 7: Enter and Exit sleep mode(UART sleep)

RTC Mode

RTC mode provides a lower current consumption than sleep mode. It is suitable for applications that remain idle for a long period. Triggering RTC mode is software configurable. The RTC timer or RESET_N can be used to exit RTC mode

Software RTC:

- Enter RTC mode:

Entering into RTC mode is controlled by UART interface, send \$PAIR650,0*25\r\n to enter RTC mode.

NOTE

For command PAIR650, please refer to “SIM32ELA Series_NMEA Message_User Guide”.

- Exit RTC Mode:

SIM32ELA module RESET_N pin can be used to exit RTC mode, when RESET_N is pulled down for 10ms, the module will exit RTC mode.

Case1: Using host control power source

Step1: enable external power source with have to meeting power on sequence.

Step2: Pull down RESET_N 10ms.

Step3: SIM32ELA module will execute the reset command and auto exit RTC mode.

Hardware RTC:

- Enter RTC mode:

Disconnect VCC. The V_BACKUP must be connected.

- Exit RTC Mode:

Reconnect VCC and V_BACKUP keep connected.

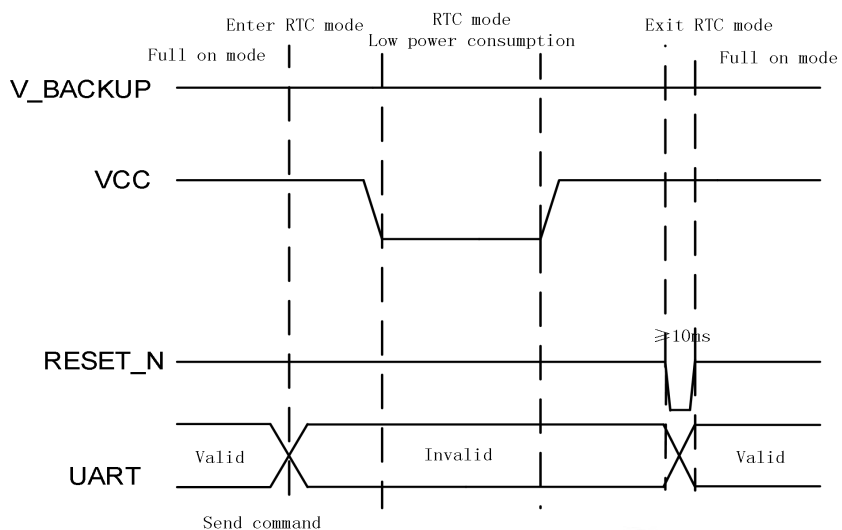


Figure 8: (Software RTC)Enter and Exit RTC Mode

3.2 WAKE_UP Signal

The WAKE_UP output high level indicating to host that the module is active. WAKE_UP output is intended to drive only CMOS inputs; do not load WAKE_UP signal with current exceeding 10mA. Only loads with steady state current drain is allowed.

3.3 ENIT_IN Signal

The module can be configured with UART sleep to reduce power consumption. ENIT_IN is used to wake up module from sleep.

If lower power consumption is required, you can send \$PAIR382,0*2F command to turn on the UART Sleep function. After turning on the serial port sleep function, \$PAIR commands must be sent in the following timing:

1. Pull down "EINT-IN" pin at least 10ms, in the process of pulling down EINT 10ms, SIM65M will return \$PAIR012*39.
2. Send command within 100ms after returning \$PAIR012*39.

NOTE

You can ignore this signal, if you do not use the module uart sleep function.

3.4 UART Interface

UART can provide the developers with signal or message outputs. SIM32ELA includes two UART interfaces.

- UART0 interface for serial communication, and this UART support NMEA output and PAIR command input.
- UART1 interface is the RTCM format data input used for RTCM function.

Table 7: Host port multiplexed function pins

Pin name	Pin number	I/O	Description	Comment
TXD0	25	O	Serial data output of NMEA	
RXD0	26	I	Serial data input for firmware update	2.8V power domain
TXD1	3	O	Default UART1	2.8V power domain, If
RXD1	6	I	RTCM function	unused, keep open.

NOTE

The default baud rate is 9600 or 115200, if other baud rate required please contact SIMCom.

The following figure shows the connection between module and client (DTE).

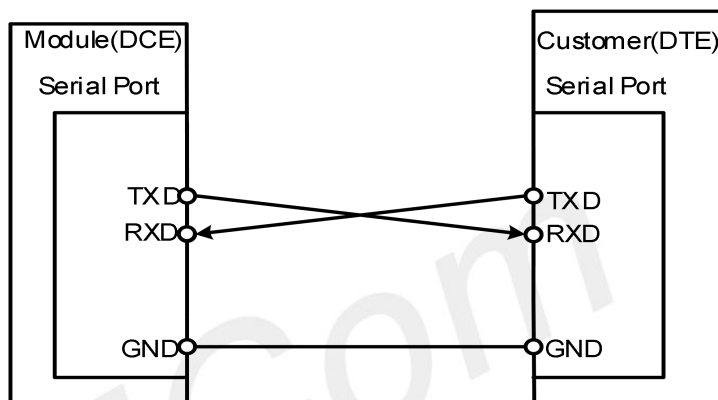


Figure 9: Connection of the serial interfaces

3.5 TIMEMARK Output

The Timemark pin outputs pulse-per-second (PPS) pulse signal for precise timing purposes after the position has been fixed. The Timemark signal can be provided through designated output pin for many external applications. This pulse is not only limited to be active every second but also allowed to set the required duration, frequency, and active high/low by programming user-defined settings.

PPS GPS time reference with adjustable duty cycle and +/- 10ns accuracy, support for time service application, which is achieved by the PPS vs NMEA feature.

The following figure is the typical application of the TIMEMARK function.

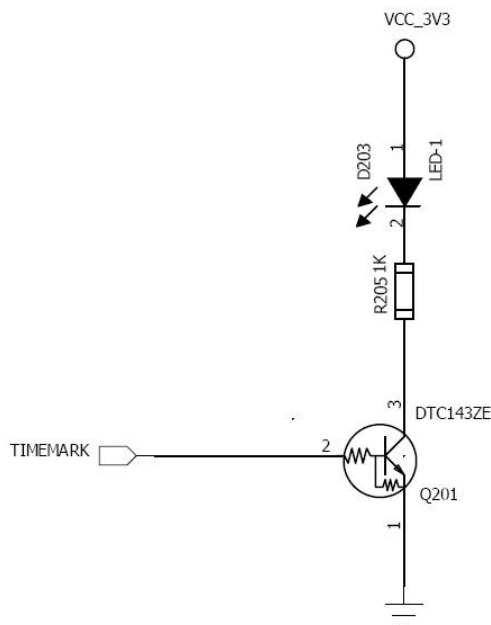


Figure 10: TIMEMARK application circuit

3.6 A-GPS

A-GPS is the meaning of Assisted GPS, which is a system that can under certain conditions improve the startup performance, or time-to-first-fix (TTFF) of a GPS satellite-based positioning system. SIM32ELA module supports EPO file, EASY mode, SBAS and RTCM.

3.6.1 EPO

The SIM32ELA supports the EPO (Extended Prediction Orbit) data service. The EPO data service is supporting 7/14/30-day orbit predictions to customers. It needs occasional download from EPO server. Supply of aiding information like ephemeris, almanac, rough last position and time and satellite status and an optional time synchronization signal will reduce time to first fix significantly.

The user should update the EPO files from the EPO server in the period of validity of EPO file through the internet. Then the EPO data should send to the SIM32ELA by the HOST side. SIM32ELA has the shorter cold TTFF and warm TTFF, when the A-GPS is used.

NOTE

For more information about EPO, please contact SIMCom.

3.6.2 EASY Mode

EASY is the abbreviation of Embedded Assist System, it works as embedded software which accelerates TTFF by predicting satellite navigation messages from received ephemeris.

No additional computing interval for EASY task. EASY is efficiently scheduled and computed in free time of every second after GPS navigation solution.

Easy function is conceptually designed to automatically engage for predicting after first receiving the broadcast ephemeris. After a while (generally tens of seconds), 3-day extensions will be completely generated then all EASY functions will be maintained at a standby condition. EASY assistance is going to be engaged when the GPS requests in new TTFF condition or re-generates again with another new received ephemeris. Meanwhile, TTFF will be benefited by EASY assistance.

NOTE

EASY function is default open and can be closed by PAIR command.

3.6.3 SBAS and RTCM

3.8.3.1 SBAS

SBAS is the abbreviation of Satellite Based Augmentation System. The SBAS concept is based on the transmission of differential corrections and integrity messages for navigation satellites that are within sight of a network of reference stations deployed across an entire continent. SBAS messages are broadcast via geostationary satellites able to cover vast areas.

Several countries have implemented their own satellite-based augmentation system. Europe has the European Geostationary Navigation Overlay Service (EGNOS) which covers Western Europe and beyond. The USA has its Wide Area Augmentation System (WAAS). Japan is covered by its Multi-functional Satellite Augmentation System (MSAS). India has launched its own SBAS program named GPS and GEO Augmented Navigation (GAGAN) to cover the Indian subcontinent.

3.8.3.2 RTCM

SIM32ELA module supports soft RTCM, but only one mode can be applied at one time, and SBAS is the default feature, customers who want to apply RTCM in the design can contact SIMCom for supporting.

3.7 Antenna

SIM32ELA has an embedded GNSS Antenna and the signal is further amplified by internal Low Noise Amplifier (LNA), which is available at ANT output. The antenna signal ANT shall be connected externally to RF_IN Antenna Input signal via a short trace between pad ANT and pad RF_IN.

3.7.1 Antenna Interface

The SIM32ELA receives L1 band signals from GPS/GLONASS/BEIDOU satellites, The RF signal is connected to the RF_IN pin. And the trace from RF_IN to antenna should be controlled to 50Ω impedance.

To suit the physical design of individual applications the RF interface pad can lead to two alternatives:

Recommended approach: solderable RF coaxial cable assembly antenna connector, such as HRS' U.FL-R-SMT (10) connector or I-PEX's 20279-001E-01 RF connector.
SMA connector.

3.7.2 Antenna Choice Consideration

To obtain excellent GNSS reception performance, a good antenna will always be required. Proper choice and placement of the antenna will ensure that satellites at all elevations can be seen, and therefore, accurate fix measurements are obtained.

The total noise figure should be well below 3 dB.

If a patch antenna is the preferred antenna, choose a patch of at least 15x15x4 mm for standalone.

GPS/QZSS, or choose a patch of at least 25x25x4 mm for GPS + GLONASS. For smaller antennas, an LNA with a noise figure <2 dB is recommended.

Make sure the antenna is not located close to noisy parts of the circuitry (e.g. micro-controller or High Power or display).

To optimize performance in environments with out-of-band jamming sources, use an additional SAW filter. The micro strip must be 50 Ω and be routed in a section of the PCB where minimal interference from noise sources can be expected.

In case of a multi-layer PCB, use the thickness of the dielectric between the signal and the first GND layer (typically the 2nd layer) for the micro strip calculation.

If the distance between the micro strip and the adjacent GND area (on the same layer) does not exceed 5 times the track width of the micro strip.

Use an external LNA if your design does not include an active antenna when optimal performance is important.

The following table shows GNSS Operating frequencies

Table 8: GNSS Operating frequencies

Type	Frequency
GPS L1	1575.42±1.023MHz
GLONASS G1	1601.7±6.75MHz
Galileo E1	1575.42±1.023MHz
BeiDou B1C	1575.42±16.368MHz
BeiDou B1I	1561.098±2.046MHz
BeiDou B2a	1176.45±10.23MHz
Galileo E5a	1176.45±10.23MHz

The suggested active antenna should be chosen as following:

Table 9: Antenna Specifications

	Specification	Passive and active antenna
Active Antenna Recommendations	Frequency range L1	1560~1609MHz
	Polarization	RHCP
	Gain	>16dB (max 35 dB)
	Noise Figure	<1.5 dB

3.7.2.1 Passive Antenna

Passive antenna contains only the radiating element, e.g. the ceramic patch, the helix structure, and chip antennas. Sometimes it also contains a passive matching network to match the electrical connection to 50 Ohms impedance.

The most common antenna type for GPS/GLONASS/BEIDOU application is the patch antenna. Patch antennas are flat, generally have a ceramic and metal body and are mounted on a metal base plate.

Figure 11 shows a minimal setup for a GPS/GLONASS/BEIDOU receiver with SIM32ELA module.

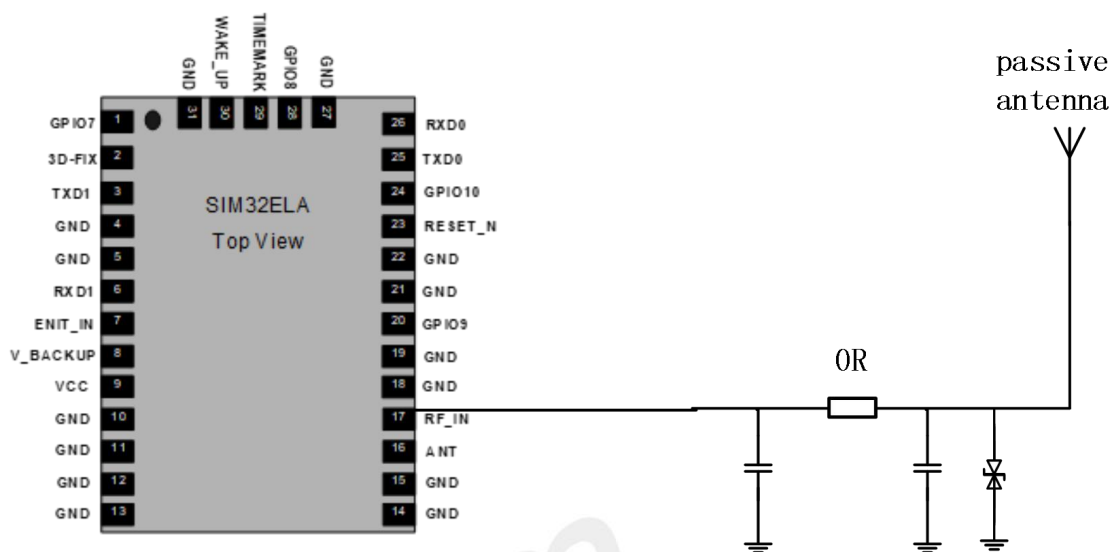


Figure 11: SIM32ELA passive antenna design

NOTE

TVS requires ultra-low junction capacitance, less than 0.05 pF is recommended.

For better performance with passive antenna designs user can use an external LNA to increase the sensitivity up 3~4 dB. Please see Figure 12.

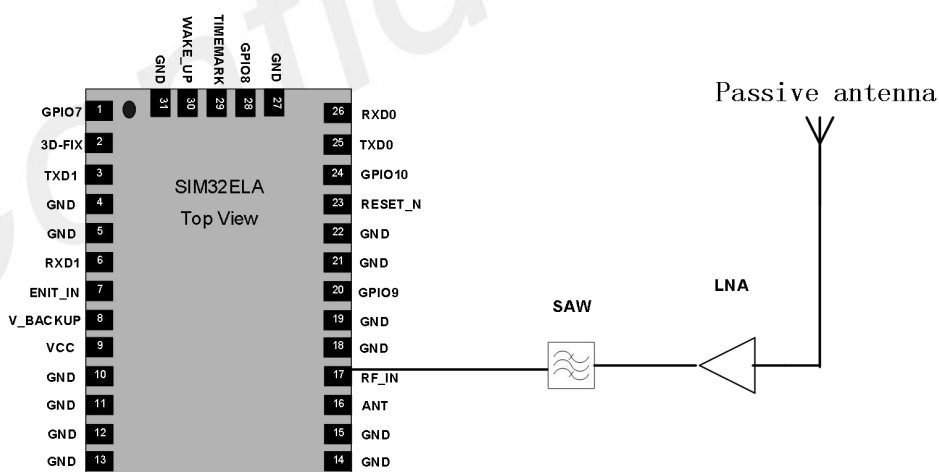


Figure 12: SIM32ELA passive antenna design (with external LNA and SAW)

For best performance, user can add an external saw based on Figure 12 design to avoid interference, please see Figure 13.

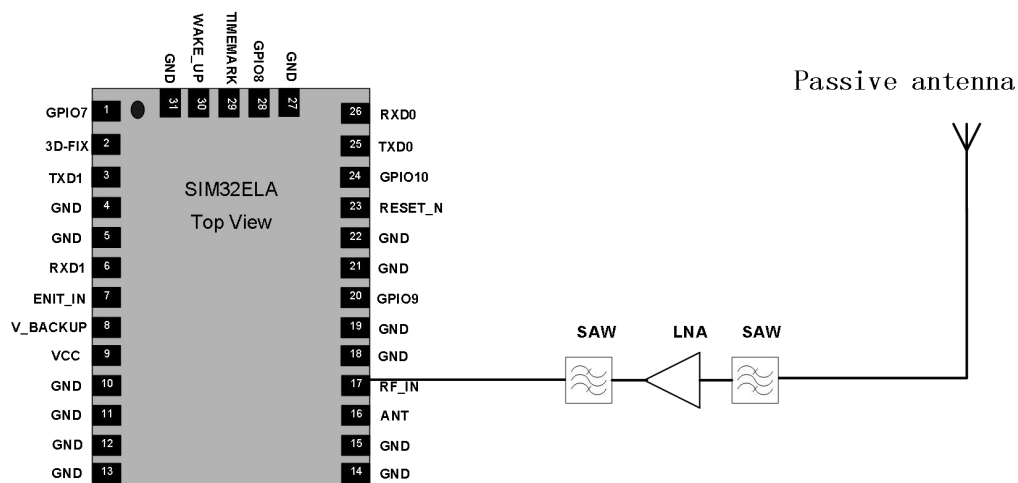


Figure 13: SIM32ELA passive antenna design

3.7.2.2 Reference Design for Antenna Embedded

If using the embedded antenna, make sure the pad ANT be connected externally to RF_IN pad via a short trace with $50\ \Omega$ impedance. The RF input signal path contains a first SAW band-pass filter before LNA, which provides excellent protection against out-of-band GNSS blocking caused by possible near-by wireless transmitters. Figure 14 shows the reference design circuit for embedded antenna.

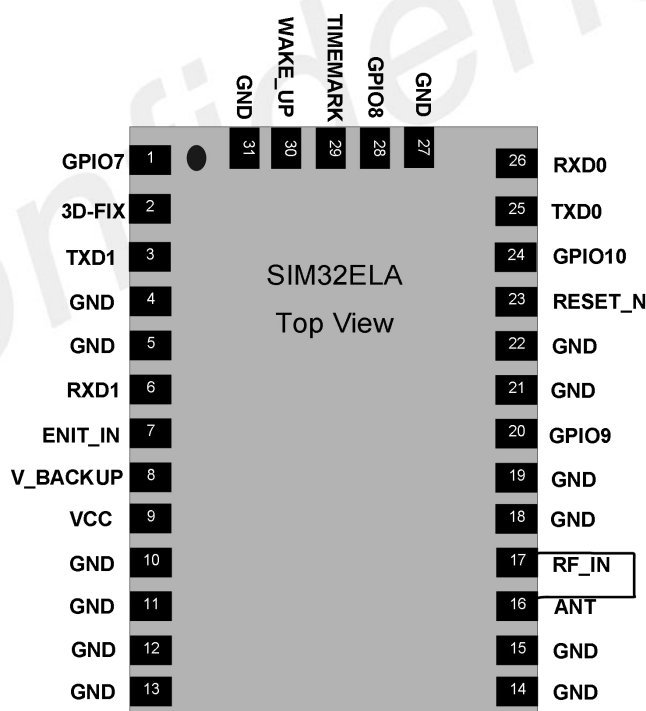


Figure 14: Reference design circuit for antenna embedded

4 Electrical Characteristics

4.1 Absolute Maximum Ratings

The absolute maximum ratings stated in Table 10 are stress ratings under non-operating conditions. Stresses beyond any of these limits will cause permanent damage to SIM32ELA.

Table 10: Absolute maximum ratings

Parameter	Min	Max	Unit
VCC	-	5.5	V
RF_IN	-	3.6	V
V_BACKUP	-	3.63	V
I/O pin voltage	-	3.63	V
Storage temperature	-45	+95	°C
Operating Temperature	-40	+85	°C

NOTE

The absolute maximum rating of RF_IN please refer to the active antenna datasheet

4.2 Recommended Operating Conditions

Table 11: SIM32ELA operating conditions

Parameter	Symbol	Min	Typ	Max	Unit
Operating temperature range		-40	+25	+85	°C
Main supply voltage	VCC	2.8	3.3	4.3	V
	I _{max}			100	mA
Module active indicate	WAKE_UP		1.8		V
Backup battery voltage	V_BACKUP	1.7	2.8	3.6	V

Table 12: SIM32ELA standard IO features

Parameter	Symbol	Min	Typ	Max	Unit
Low level output voltage Test conditions IOL = 2mA and 4.0mA@2.8V	V _{ol}	-0.3		0.35	V
High level output voltage Test conditions IOL = 2mA and 4.0mA@2.8V	V _{oh}	2.1		3.1	V
Low level input voltage@2.8V	V _{il}	-0.3		0.7	V
High level input voltage@2.8V	V _{ih}	1.75		3.1	V
Low level output voltage Test conditions IOL = 2mA and 4.0mA@1.8V	V _{ol}	-0.3		0.45	V
High level output voltage Test conditions IOL = 2mA and 4.0mA@1.8V	V _{oh}	1.35		1.98	V
Low level input voltage@1.8V	V _{il}	-0.3		0.63	V
High level input voltage@1.8V	V _{ih}	1.17		1.98	V
Input Pull-up resistance	RPU	40	75	190	K Ω
Input Pull-down resistance	RPD	40	75	190	K Ω
Input capacitance	Cin		5		pF
Load capacitance	Cload			8	pF
Tri-state leakage current	IOZ	-10		10	uA

4.3 Electro-Static Discharge

The GPS engine is not protected against Electrostatic Discharge (ESD) in general. Therefore, it is subject to ESD handling precautions that typically apply to ESD sensitive components. Proper ESD handling and packaging procedures must be applied throughout the processing, handling and operation of any application using a SIM32ELA module.

Table 13: The ESD characteristics (Temperature: 25°C, Humidity: 45 %)

Pin	Contact discharge	Air discharge
VCC	± 4 kV	± 8 kV
GND	± 4 kV	± 8 kV
RF_IN	± 4 kV	± 8 kV

NOTE

Test conditions:

1. The external of the module has surge protection diodes and ESD protection diodes.
2. The data in Table 13 were tested using SIMCom EVB.

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5 Manufacturing

5.1 Top and bottom View of SIM32ELA

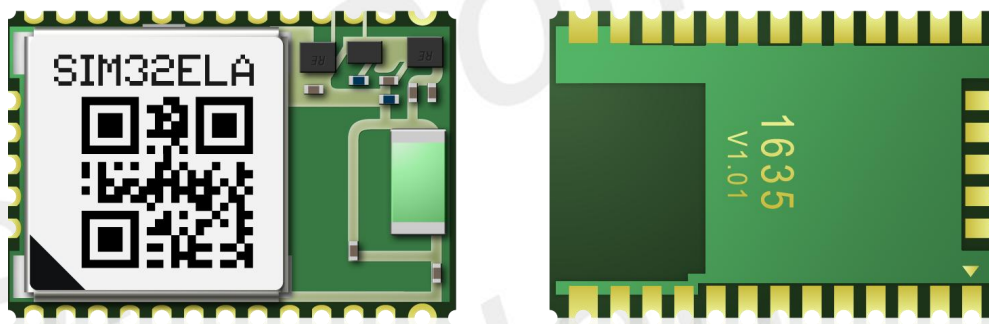


Figure 15: Top and bottom view of SIM32ELA

NOTE

The above is the design effect diagram of the module for reference. The actual appearance is subject to the actual product.

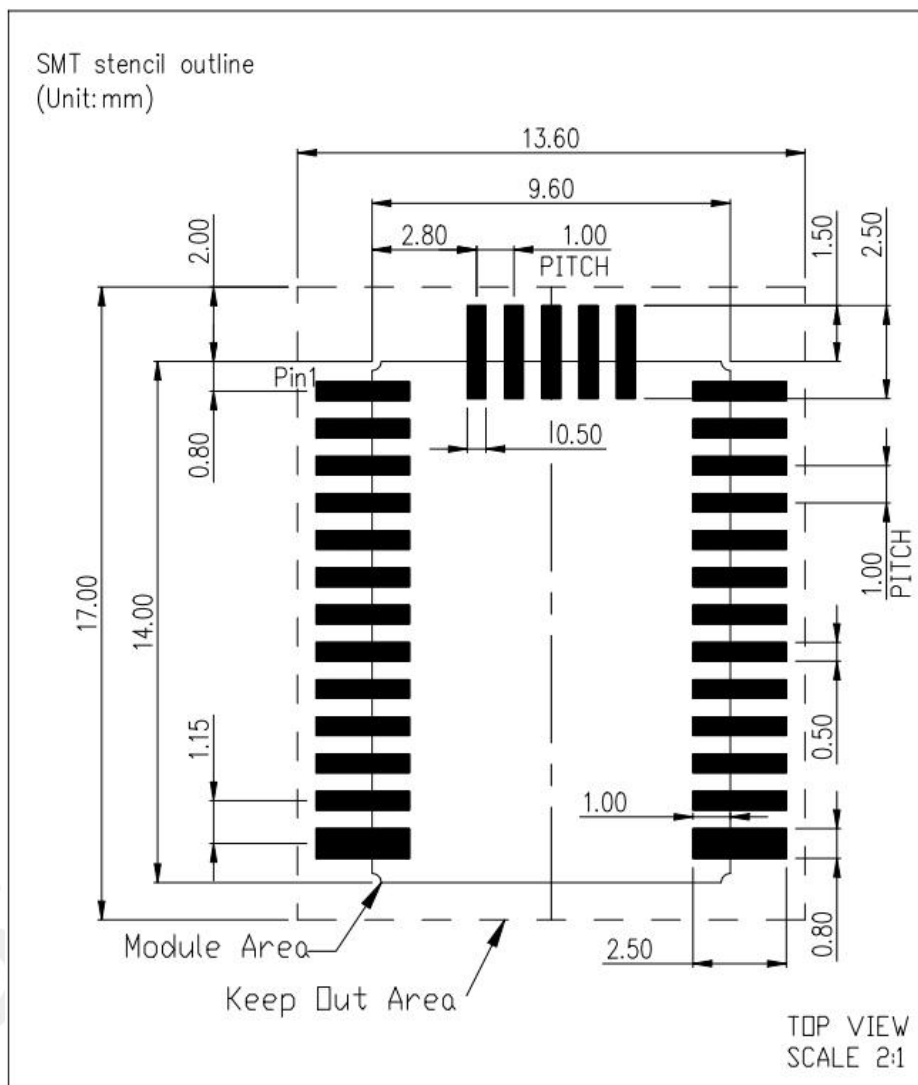


Figure 17: Recommended SMT stencil

5.4 Assembly and Soldering

The SIM32ELA module is intended for SMT assembly and soldering in a Pb-free reflow process on the top side of the PCB. Suggested solder paste stencil height is 150um minimum to ensure sufficient solder volume. If required paste mask pad openings can be increased to ensure proper soldering and solder wetting over pads.

The following figure is the Ramp-Soak-Spike Reflow Profile of SIM32ELA:

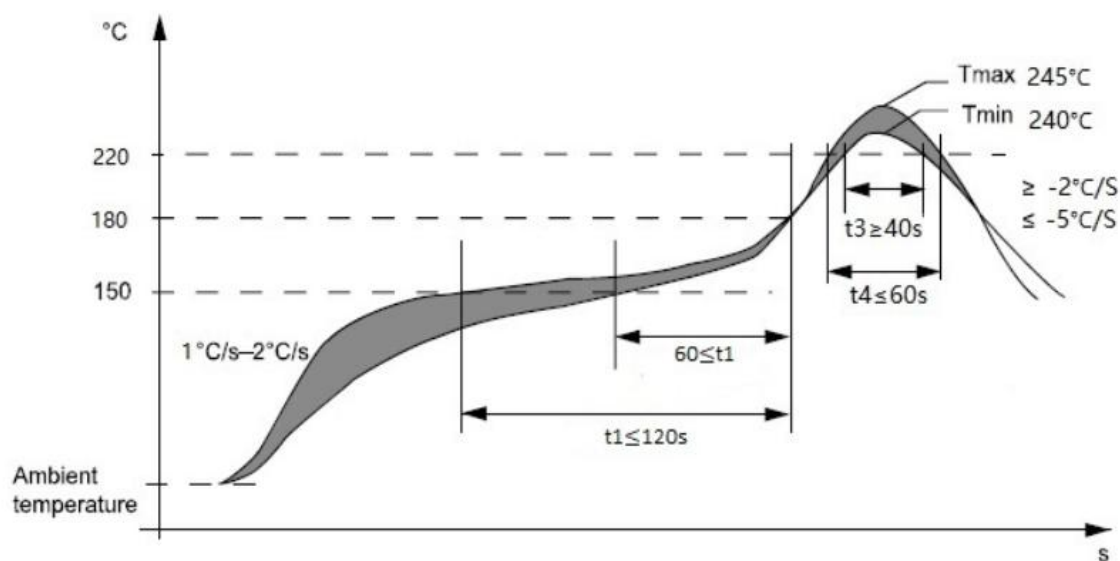


Figure 18: The Ramp-Soak-Spike reflow profile of SIM32ELA

SIM32ELA is Moisture Sensitive Devices (MSD), appropriate MSD handling instruction and precautions are summarized in Chapter 5.5.

SIM32ELA modules are also Electrostatic Sensitive Devices (ESD), handling SIM32ELA modules without proper ESD protection may destroy or damage them permanently.

Avoid ultrasonic exposure due to internal crystal and SAW components.

5.5 Moisture sensitivity

SIM32ELA module is moisture sensitive at MSL level 3, dry packed according to IPC/JEDEC specification J-STD-020C. The calculated shelf life for dry packed SMD packages is a minimum of 12 months from the bag seal date, when stored in a non condensing atmospheric environment of <40°C/90% RH.

Table 14 lists floor life for different MSL levels in the IPC/JDEC specification:

Table 14: Moisture Classification Level and Floor Life

Level	Floor Life(out of bag)at factory ambient $\leq +30^{\circ}\text{C}/60\%\text{RH}$ or as stated
1	Unlimited at $\leq +30^{\circ}\text{C}/85\%\text{RH}$
2	1 year
2a	4 weeks
3	168 hours
4	72 hours

5	48 hours
5a	24 hours
6	Mandatory bake before use. After bake, module must be reflowed within the time limit specified on the label.

Factory floor life is 1 week for MSL 3, SIM32ELA must be processed and soldered within the time. If this time is exceeded, or the humidity indicator card in the sealed package indicates that they have been exposed to moisture, the devices need to be pre-baked before the reflow solder process.

Both encapsulate and substrate materials absorb moisture. IPC/JEDEC specification J-STD-020 must be observed to prevent cracking and delamination associated with the "popcorn" effect during reflow soldering. The popcorn effect can be described as miniature explosions of evaporating moisture. Baking before processing is required in the following cases:

Humidity indicator card: At least one circular indicator is no longer blue

Floor life or environmental requirements after opening the seal have been exceeded, e.g. exposure to excessive seasonal humidity.

Refer to Section 4 of IPC/JEDEC J-STD-033 for recommended baking procedures.

NOTE

Oxidation Risk: Baking SMD packages may cause oxidation and/or inter metallic growth of the terminations, which if excessive can result in solder ability problems during board assembly. The temperature and time for baking SMD packages are therefore limited by solder ability considerations. The cumulative bake time at a temperature greater than 90°C and up to 125°C shall not exceed 96 hours.

5.6 ESD handling precautions

SIM32ELA modules are Electrostatic Sensitive Devices (ESD). Observe precautions for handling!



Failure to observe these precautions can result in severe damage to the GPS receiver!

GPS receivers are Electrostatic Sensitive Devices (ESD) and require special precautions when handling. Particular care must be exercised when handling patch antennas, due to the risk of electrostatic charges. In addition to standard ESD safety practices, the following measures should be taken into account whenever handling the receiver:

Unless there is a galvanic coupling between the local GND (i.e. the work Table) and the PCB GND, then the first point of contact when handling the PCB shall always be between the local GND and PCB GND. Before mounting an antenna patch, connect ground of the device

When handling the RF pin, do not come into contact with any charged capacitors and be careful when contacting materials that can develop charges (e.g. patch antenna ~10pF, coax cable ~50-80pF/m, soldering iron,)

To prevent electrostatic discharge through the RF input, do not touch the mounted patch antenna.

When soldering RF connectors and patch antennas to the receiver's RF pin, the user must make sure to use an ESD safe soldering iron (tip).

5.7 Shipment

SIM32ELA is designed and packaged to be processed in an automatic assembly line, and it is now packaged in SIM32ELA tray.

6 Reference Design

The figure below is a typical application of SIM32ELA using internal passive antenna.

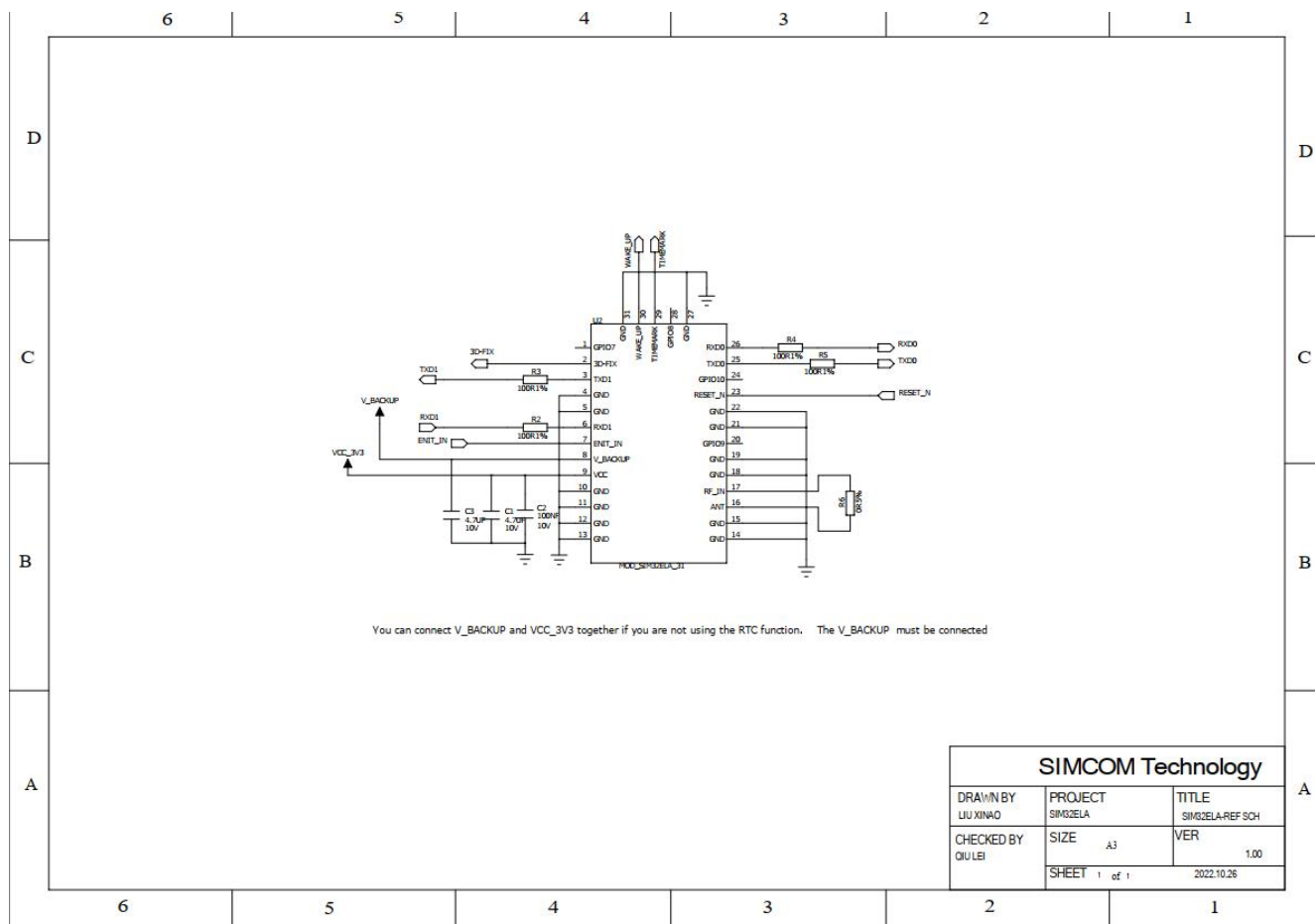


Figure 19: Refer schematic

NOTE

- 1、I/Os of SIM32ELA are 2.8V and 1.8V CMOS voltage level; attentions should be paid if the voltage level of the host controller not compatible.

7 Appendix

7.1 Related Documents

Table 15: Related documents

SN	Document name	Remark
[1]	SIM32ELA Series_NMEA Message_User Guide	
[2]	SIM32ELA REFERENCE DESIGN	

7.2 Terms and Abbreviations

Table 16: Terms and abbreviations

Abbreviation	Description
A-GPS	Assisted- Global Positioning System
CMOS	Complementary Metal Oxide Semiconductor
DGPS	Difference Global Positioning System
EASY	Embedded Assist System
EEPROM	Electrically Erasable Programmable Read Only Memory
EGNOS	Euro Geostationary Navigation Overlay Service
EPO	Extended Prediction Orbit
ESD	Electrostatic Sensitive Devices
FSM	Finite State Machine
GAGAN	The GPS Aided Geo Augmented Navigation
GPS	Global Positioning System
GNSS	Global Navigation Satellite System
I/O	Input/Output
IC	Integrated Circuit
Inorm	Normal Current
Imax	Maximum Load Current
kbps	Kilo bits per second
LNA	Low Noise Amplifier
MSAS	Multi-Functional Satellite Augmentation
MSL	moisture sensitive level

NMEA	National Marine Electronics Association
QZSS	Quasi-Zenith Satellites System
RTCM	Radio Technical Commission for Maritime Services
SBAS	Satellite Based Augmentation Systems
WAAS	Wide Area Augmentation System

7.3 Safety Caution

Table 17: Safety caution

Marks	Requirements
	When in a hospital or other health care facility, observe the restrictions about the use of mobiles. Switch the cellular terminal or mobile off, medical equipment may be sensitive and not operate normally due to RF energy interference.
	Switch off the cellular terminal or mobile before boarding an aircraft. Make sure it is switched off. The operation of wireless appliances in an aircraft is forbidden to prevent interference with communication systems. Forgetting to think much of these instructions may impact the flight safety, or offend local legal action, or both.
	Do not operate the cellular terminal or mobile in the presence of flammable gases or fumes. Switch off the cellular terminal when you are near petrol stations, fuel depots, chemical plants or where blasting operations are in progress. Operation of any electrical equipment in potentially explosive atmospheres can constitute a safety hazard.
	Your cellular terminal or mobile receives and transmits radio frequency energy while switched on. RF interference can occur if it is used close to TV sets, radios, computers or other electric equipment.
	Road safety comes first! Do not use a hand-held cellular terminal or mobile when driving a vehicle, unless it is securely mounted in a holder for hands free operation. Before making a call with a hand-held terminal or mobile, park the vehicle.
	Mobiles operate over radio frequency signals and cellular networks and cannot be guaranteed to connect in all conditions, especially with a mobile fee or an invalid (U)SIM card. While you are in this condition and need emergent help, please remember to use emergency calls. In order to make or receive calls, the cellular terminal or mobile must be switched on and in a service area with adequate cellular signal strength. Some networks do not allow for emergency call if certain network services or phone features are in use (e.g. lock functions, fixed dialing etc.). You may have to deactivate those features before you can make an emergency call. Also, some networks require that a valid (U)SIM card be properly inserted in the cellular terminal or mobile.